Physics from first principles

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Euclidean volume and some distance equations are proved to be instances of ordered combinations (n-tuples) of elements from a list of disjoint sets. Volume has a permutation property, where that permutation property is proved to limit a set of unordered elements to at most 3 elements: for example, a set of 3 unordered distance dimensions, {width, height, depth}, and, for example, a set of 3 unordered non-distance dimensions, {time, mass, charge}, each dimension a subset of R. Dividing the dimensions of Euclidean distance, time, mass, and charge into the same number of intervals, there are 3 direct and 3 inverse proportion ratios of a i-th interval length of Euclidean distance to the i-th interval lengths of time, mass, and charge. Derivations of well-known gravity, charge, electromagnetic, relativity, quantum physics equations and related constants from the ratios are shorter and simpler. Inverse ratios are used to add quantum extensions to the Schwarzchild metric, Newton's gravity force, and Coulomb's charge force equations, which predict that: 1) The gravity and charge constants do not exist where the quantum effects are measurable; 2) The force between two point masses and between two point charges peaks at the Planck unit length. The math proofs are verified in Rocq.

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I. INTRODUCTION

A. Insufficient reduction

Much of physics relies on volume and distance equations. Mathemtical analysis defines Euclidean volume as the product of interval lengths and defines the Euclidean distance equation, dot product, inner product, and metric space.[2,3]. Justification for the definitions is vigorous finger-pointing to Euclidean and Cartesian geometry.

Further, the trend in physics has been increasingly complicated equations, assumptions, and mathematics, with increasing difficulty in finding exact solutions to the equations. Examples are the Einstein field equations (general theory of relativity)[4,5] and Schrödenger's wave equation (quantum physics).[6]

And physics equations often rely on assumptions that have not had a mathematical justification. For example, relativity and quantum theory both assume:

1. 3 dimensions of space. But integration, Hilbert spaces, linear algebra, (pseudo-)Riemann manifolds, and so on, have no restriction on the number of dimensions.[2,3,5,7,8]

2. The laws of physics are the same at every local coordinate point[4–6,9] But no first principle properties have been identified that require the same functions at every local coordinate point.

The philosophy of reductionism posits that complex phenomena can be reduced to a few simple objects and relations. The first reduction step, in this article, proves that well-known volume and distance equations are reduced to (derived from) a few simple set and number properties. The second step is to identify two salient properties of the set properties and corresponding volume and distance properties. The third reduction step is to show that many well-known physics equations, constants, and assumptions are reduced to (derived from) those two properties.

B. First reduction step

If the underlying principles are simple, then the proofs that well-known geometry equations are reduced to (derived from) a few simple set relations and the properties of $\mathbb{R}$ should be simple, trivial even. But even apparently trivial proofs can have subtle flaws.

Therefore, each math proof, in this article, has a corresponding formal proof, which has been verified using using the Rocq proof verification system,[10] a verification tool used by mathematicians world-wide. The formal proofs are in the Rocq files, "euclidrelations.v" and "threed.v," which are included as ancillary files.

Let $|x_i|$ be the cardinal of (number of elements in) the abstract, countable set, $x_i \in \{x_1, \cdots, x_n\}$. And let the value, v_c, be the number of ordered combinations (n-tuples) of the elements of $x_1, \cdots, x_n$:

$$x_i \in \{x_1, \cdots, x_n\}, \quad \bigcap_{i=1}^n x_i = \emptyset, \quad |x_i| \in \{0, \mathbb{N}\} \quad \wedge \quad v_c = \prod_{i=1}^n |x_i|. \tag{1}$$

Where each n-tuple corresponds to a scalar value, $\kappa \in \mathbb{R}$, it will be proved that:

$$v_c = \prod_{i=1}^n |x_i| \quad \Rightarrow \quad \exists\, v, s_i \in \mathbb{R}: \;\; v = \prod_{i=1}^n s_i. \tag{2}$$

For all $n > 1$, there are cases where different domain values, $|x_1|, \cdots, |x_n|$, multiplied yield the same range value, v_c. Inferring a domain value, d_c, from v_c, requires an inverse (bijective) function, $d_c = f_n^{-1}(v_c)$ and $v_c = f_n(d_c)$. The simplest bijective case for all n is the cuboid case, $|x_i| = d_c$. And it will be proved that:

$$v_c = \prod_{i=1}^n |x_i| \quad \wedge \quad d_c = |x_i| \quad \Rightarrow \quad \exists\, v, d \in \mathbb{R}: \;\; v = d^n. \tag{3}$$

Where v_c is the sum of the numbers of subset n-tuples, it will be proved that:

$$d_c^n = v_c = \sum_{i=1}^m v_{c_i} \quad \Rightarrow \quad d^n = v = \sum_{i=1}^m \left(\prod_{j=1}^n s_{i,j}\right). \tag{4}$$

Note that volume, v, is a scalar quantity, where the $n = 2$ case is the dot product:

$$v = \sum_{i=1}^{m} (\prod_{j=1}^{n=2} s_{i,j}) \ \wedge \ \exists\, a_i, b_i \in \mathbb{R} : \ \ s_{i,1} = a_i, \ \ s_{i,2} = b_i \ \ \Rightarrow \ \ v = \sum_{i=1}^{m} a_i b_i \stackrel{\text{def}}{=} \mathbf{a} \cdot \mathbf{b}. \quad (5)$$

The inner product is where a_i and b_i are allowed to also be complex numbers, functions, and matrices.

And, where each summed volume, v_{c_i}, is also the bijective function, $d_{c_i}^n = v_{c_i}$:

$$d_c^n = v_c = \sum_{i=1}^{m} v_{c_i} = \sum_{i=1}^{m} d_{c_i}^n \quad \Rightarrow \quad d^n = \sum_{i=1}^{m} d_i^n. \quad (6)$$

$|d|$ is the p-norm (Minkowski distance),[11] which will be proved to imply the metric space properties.[3] The $n = 2$ case is the Euclidean distance.

C. Second reduction step

For all Euclidean and non-Euclidean volumes, $v \in \mathbb{R}$, there exists $d \in \mathbb{R}$, such that $d = v^{1/n}$ and $d^n = v$. The consequence is that the dimension of all Euclidean distances $\subseteq \mathbb{R}$.

The commutative property of the union of subsets of n-tuples and the summing or multiplication operations defining the number n-tuples and corresponding Euclidean volume (2) allow n number of domain elements to be sequenced in all $n!$ permutations, for example, $v = \mathit{width} \times \mathit{height} \times \mathit{depth} = \mathit{depth} \times \mathit{width} \times \mathit{height} = \cdots$. The commutative property implies that each domain element has no intrinsic property that makes an element the first, second, ... , or last, which is a set of unordered elements.

Reliably sequencing a set of unordered elements in the same order requires that the elements be placed into a "list," which assigns relative sequential positions, for example, labeling the unordered elements as x_1, x_2, $\cdots x_n$. The list can be sequenced via the successor and predecessor functions, $\mathit{successor}(x_i) = x_{i+1}$ and $\mathit{predecessor}(x_i) = x_{i-1}$. For example, assigning s_1 = width, s_2 = height, and s_3 = depth, the successor sequence, $v = \prod_{i=1}^{3} s_i$, reliably sequences the unordered set elements in the order, width→height→depth.

Only a cyclic list, where $\mathit{successor}(x_n) = x_1$ and $\mathit{predecessor}(x_1) = x_n$, allows each unordered element to be the first element of some of those $n!$ permutations. For example, where $n = 3$, the volume cyclic successor sequences are: $v = s_1 \times s_2 \times s_3 = s_2 \times s_3 \times s_1$ $= s_3 \times s_1 \times s_2$, and the cyclic predecessor sequences are: $v = s_1 \times s_3 \times s_2 = s_2 \times s_1 \times s_3$ $= s_3 \times s_2 \times s_1$.

The only cyclic lists that allows sequencing a set of unordered elements in all $n!$ permutations, is where each list element is cyclic sequentially adjacent (either an *immediate* cyclic successor or an *immediate* cyclic predecessor) to every other element, referred to, in this article, as an immediate symmetric cyclic list (ISCL). For example, the cyclic permutations, $[x_2, x_3, x_1]$ and $[x_2, x_1, x_3]$, require x_1 and x_3 be cyclic sequentially adjacent. An ISCL will be proved to contain $n \leq 3$ elements.

Therefore, the consequence of the ISCL property of volume is a set of at most 3 unordered distance dimensions, {width, height, depth}, each distance dimension having the $\subseteq \mathbb{R}$ membership property, and a set of at most 3 unordered non-distance dimensions $\{x, y, z\}$, each non-distance dimension having the $\subseteq \mathbb{R}$ membership property.

D. Third reduction step

This section is an overview of the third step of reducing many well-known physics equations, constants, and assumptions to the Euclidean distance dimension $\subseteq \mathbb{R}$ property and the ISCL property:

1. **ISCL:** $\{r_1, r_2, r_3\}$ is an ISCL of 3 unordered "distance" dimensions, each dimension having the $\subseteq \mathbb{R}$ set membership property and $\{t\ (\textit{time}), m\ (\textit{mass}),\ q\ (\textit{charge})\}$ is the ISCL of 3 unordered "non-distance" dimensions, each dimension $\subseteq \mathbb{R}$.

2. **Corresponding lengths:** Where the dimensions of Euclidean distance, time, mass, and charge are each divided into the same number of intervals, the i^{th} unit interval length, r_p of Euclidean distance, corresponds to the i^{th} unit interval lengths: t_p of time, m_p of mass, and q_p of charge, where "unit" means a measurement type, for example, meters, seconds, kilograms, and Coulombs.

The correspondence of unit interval lengths implies 3 direct proportion unit ratios: $r/t = r_p/t_p$, $r/m = r_p/m_p$, and $r/q = r_p/q_p$, letting: $c_t = r_p/t_p$, $c_m = r_p/m_p$, and $c_q = r_p/q_p$. And the correspondence of unit interval lengths implies 3 inverse proportion unit ratios: $rt = t_p r_p$, $rm = m_p r_p$. and $rq = q_p r_p$, And let: $k_t = t_p r_p$, $k_m = m_p r_p$, and $k_q = q_p r_p$.

The gravity,[12,13] special relativity,[4,5] Schwarzschild black hole metric,[14,15] charge equations,[13], electromagnetism,[13] and related constants are derived from the 3 direct proportion unit ratios. The Planck-Einstein relation,[13] Compton wavelength,[13], Schrödenger wave equation,[6] Dirac wave equation,[9] and related constants are derived from combining some the direct and inverse proportion unit ratios.

Determining the values of the quantum units: r_p, t_p, m_p, and q_p are enabled by the property:

$$r_p = \sqrt{r_p^2} = \sqrt{c_t k_t} = \sqrt{c_m k_m} = \sqrt{c_q k_q}. \tag{7}$$

The Planck-Einstein relation, will be derived as $mc^2 = k_m c/t$, (where the speed of light, $c = c_t$). Where the reduced Planck constant, $\hbar = k_m c$, the quantum units: r_p, t_p, m_p, and q_p will be shown to be Planck units.

The inverse proportion ratios are used to add quantum extensions to Scharwzchild's metric, Newton's gravity force, and Coulomb's charge force. The quantum-extended gravity, charge, and relativity equations predict that where quantum effects are measurable, the constants: gravity, G, Einstein's $\kappa = 8\pi G/c^4$, charge, k_e, permittivity, ε_0, and permeability, μ_0, do not exist. And the quantum-extended equations also predict that the force between two point masses and between two point charges peak at the Planck unit length.

II. VOLUME

A. Euclidean volume

Theorem II.1. *Euclidean volume,*

$$\forall\, v_c, d_c, |x_i| \in \{0,\ \mathbb{N}\},\ \ x_i \in \{x_1, \cdots, x_n\},\ v_c = \textstyle\prod_{i=1}^n |x_i| \ \Rightarrow\ v = \prod_{i=1}^n s_i, \quad s_i, v \in \mathbb{R}. \tag{8}$$

The formal proof, "Euclidean_volume," is in the Rocq file, euclidrelations.v.

Proof.

$$\forall v_c, |x_i| \in \{0, \mathbb{N}\},\ \ \kappa \in \mathbb{R}:\ \ v_c = \textstyle\prod_{i=1}^n |x_i| \ \Leftrightarrow\ v_c\kappa = (\prod_{i=1}^n |x_i|)\kappa = \prod_{i=1}^n (|x_i|\kappa^{1/n}). \tag{9}$$

$$v_c\kappa = \textstyle\prod_{i=1}^n (|x_i|\kappa^{1/n}) \ \wedge\ \exists\, v, s_i \in \mathbb{R}:\ \ v = v_c\kappa \ \wedge\ s_i = |x_i|\kappa^{1/n} \ \Rightarrow\ v = \prod_{i=1}^n s_i. \quad \square \tag{10}$$

Lemma II.2. *The number of n-tuples, v_c, is the sum of the number of n-tuples, v_{c_i}, in each disjoint subset of n-tuples, implies a volume is the sum of volumes,*

$$v_c = \textstyle\sum_{i=1}^m v_{c_i} \quad \Rightarrow \quad v = \sum_{i=1}^m v_i, \quad v, v_i \in \mathbb{R}.$$

The formal proof, "sum_of _volumes," is in the Rocq file, euclidrelations.v.

Proof. From the condition of this theorem and the Euclidean volume theorem (II.1):

$$\forall\, \kappa \in \mathbb{R}: \;\; v_c = \textstyle\sum_{i=1}^m v_{c_i} \quad\Rightarrow\quad v_c\kappa = (\sum_{i=1}^m v_{c_i})\kappa = \sum_{i=1}^m (v_{c_i}\kappa). \tag{11}$$

$$v_c\kappa = \textstyle\sum_{i=1}^m (v_{c_i}\kappa) \quad\wedge\quad v = v_c\kappa \quad\wedge\quad v_i = v_{c_i}\kappa \quad\Rightarrow\quad v = \sum_{i=1}^m v_i. \quad \square \tag{12}$$

III. DISTANCE

Definition III.1. Bijective, countable domain value, d_c:

$$\exists\, v_c, d_c, |x_i| \in \{0,\ \mathbb{N}\},\;\; x_i \in \{x_1, \cdots, x_n\}, \quad v_c = \textstyle\prod_{i=1}^n |x_i| = \prod_{i=1}^n d_c = d_c^n. \tag{13}$$

A. Vector inner product

Theorem III.2. *Sum of volumes distance:*

$$d_c^n = v_c = \textstyle\sum_{i=1}^m v_{c_i} \quad\Rightarrow\quad d^n = v = \sum_{i=1}^m (\prod_{j=1}^n s_{i_j}).$$

The formal proof, "sum_of _volumes _distance," is in the Rocq file, euclidrelations.v.

Proof. Multiply both sides of the assumption by κ and apply the Euclidean volume proof (II.1) to v_i:

$$v_c = \textstyle\sum_{i=1}^m v_{c_i} \quad\Rightarrow\quad v = \sum_{i=1}^m v_i, \quad v, v_i \in \mathbb{R}. \tag{14}$$

$$v = \textstyle\sum_{i=1}^m v_i \quad\wedge\quad v_i = \prod_j^n s_{i,j} \quad\Rightarrow\quad v = \sum_{i=1}^m (\prod_{j=1}^n s_{i,j}). \tag{15}$$

$$d_c^n = v_c \quad\Rightarrow\quad d_c^n\kappa = (d_c\kappa^{1/n})^n = v_c\kappa. \tag{16}$$

$$\exists\, d, v \in \mathbb{R} : d = d_c\kappa^{1/n} \quad\wedge\quad v = v_c\kappa \quad\wedge\quad (d_c\kappa^{1/n})^n = v_c\kappa \quad\Rightarrow\quad d^n = v. \tag{17}$$

$$d^n = v \quad\wedge\quad v = \textstyle\sum_{i=1}^m (\prod_{j=1}^n s_{i,j}) \quad\Rightarrow\quad d^n = v = \sum_{i=1}^m (\prod_{j=1}^n s_{i,j}). \quad \square \tag{18}$$

Volume, v, is a scalar quantity, where the $n = 2$ case is the dot product:

$$v = \textstyle\sum_{i=1}^m (\prod_{j=1}^{n=2} s_{i,j}) \;\wedge\; \exists\, a_{i,1}, a_{i,2} \in \mathbb{R}: \;\; s_{i,1} = a_i,\;\; s_{i,2} = b_i \;\Rightarrow\; v = \sum_{i=1}^m a_i b_i \stackrel{\text{def}}{=} \mathbf{a}\cdot\mathbf{b}. \tag{19}$$

The inner product is where a_i and b_i are allowed to also be complex numbers, functions, and matrices.

B. Minkowski distance (p-norm)

Theorem III.3. *Minkowski distance (p-norm):*

$$d_c^n = v_c = \sum_{i=1}^m v_{c_i} = \sum_{i=1}^m d_{c_i}^n \quad \Leftrightarrow \quad d^n = \sum_{i=1}^m d_i^n.$$

The formal proof, "Minkowski_distance," is in the Rocq file, euclidrelations.v.

Proof. From the sum of volumes distance theorem III.2:

$$d_{c_i}^n = v_{c_i} \quad \Rightarrow \quad d_i^n = v_i. \tag{20}$$

$$d^n = v = \sum_{i=1}^m v_i \quad \wedge \quad d_i^n = v_i \Rightarrow \quad d^n = \sum_{i=1}^m d_i^n \quad \square \tag{21}$$

C. Distance inequalities

Theorem III.4. *Distance triangle inequality*

$$\forall\, n \in \mathbb{N},\;\; v_a, v_b \geq 0 : \;\; (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}.$$

The formal proof, distance_inequality, is in the Rocq file, euclidrelations.v.

Proof. Expand $(v_a^{1/n} + v_b^{1/n})^n$ using the binomial expansion:

$$\forall\, v_a, v_b \geq 0 : \quad v_a + v_b \leq v_a + v_b + \sum_{i=1}^n \binom{n}{k} (v_a^{1/n})^{n-k} (v_b^{1/n})^k + \sum_{i=1}^n \binom{n}{k} (v_a^{1/n})^k (v_b^{1/n})^{n-k} = (v_a^{1/n} + v_b^{1/n})^n. \tag{22}$$

Take the n^{th} root of both sides of the inequality 22:

$$\forall\, v_a, v_b \geq 0,\;\; n \in \mathbb{N} : \quad v_a + v_b \leq (v_a^{1/n} + v_b^{1/n})^n \quad \Rightarrow \quad (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}. \tag{23}$$

$\square$

Theorem III.5. *Distance sum inequality*

$$\forall\, m, n \in \mathbb{N},\;\; a_i, b_i \geq 0 : (\textstyle\sum_{i=1}^m (a_i^n + b_i^n))^{1/n} \leq (\sum_{i=1}^m a_i^n)^{1/n} + (\sum_{i=1}^m b_i^n)^{1/n}.$$

The formal proof, distance_sum_inequality, is in the Rocq file, euclidrelations.v.

Proof. Apply the distance triangle inequality (III.4):

$$\forall\, m, n \in \mathbb{N},\;\; v_a, v_b \geq 0 : \;\; v_a = \sum_{i=1}^m a_i^n \quad \wedge \quad v_b = \sum_{i=1}^m b_i^n \quad \wedge \quad (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}$$
$$\Rightarrow \quad ((\textstyle\sum_{i=1}^m a_i^n) + (\sum_{i=1}^m b_i^n))^{1/n} = (\sum_{i=1}^m (a_i^n + b_i^n))^{1/n} \leq (\sum_{i=1}^m a_i^n)^{1/n} + (\sum_{i=1}^m b_i^n)^{1/n}. \quad \square \tag{24}$$

D. Metric Space

All Minkowski distances (p-norms) imply the metric space properties. The formal proofs: triangle_inequality, symmetry, non_negativity, and identity_of_indiscernibles are in the Rocq file, euclidrelations.v.

Theorem III.6. *Triangle Inequality:*

$$d(s_1, s_2) = (\textstyle\sum_{i=1}^{2} s_i^p)^{1/p}, \quad p \geq 1, \quad u = s_1, \quad w = s_2, \quad v = w/k$$
$$\Rightarrow \quad d(u, w) = (u^p + w^p)^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p} = d(u, v) + d(v, w).$$

Proof. $\forall\, p \geq 1, \quad u = s_1, \quad w = s_2, \quad v = w/k$:

$$(u^p + w^p)^{1/p} \leq ((u^p + w^p) + 2v^p)^{1/p} = ((u^p + v^p) + (v^p + w^p))^{1/p}. \tag{25}$$

Apply the distance triangle inequality (III.4) to the inequality 25:

$$(u^p + w^p)^{1/p} \leq ((u^p + v^p) + (v^p + w^p))^{1/p} \quad \wedge \quad (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}$$
$$\wedge \quad v_a = u^p + v^p \quad \wedge \quad v_b = v^p + w^p$$
$$\Rightarrow \quad (u^p + w^p)^{1/p} \leq ((u^p + v^p) + (v^p + w^p))^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p}$$
$$\Rightarrow \quad d(u, w) = (u^p + w^p)^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p} = d(u, v) + d(v, w). \quad \square \tag{26}$$

Theorem III.7. *Symmetry:*
$d(s_1, s_2) = (\sum_{i=1}^{2} s_i^p)^{1/p}, \quad p \geq 1, \quad u = s_1, \quad v = s_2 \quad \Rightarrow \quad d(u, v) = d(v, u).$

Proof. By the commutative law of addition:

$$\forall\, p \;:\; p \geq 1, \quad d(s_1, s_2) = (\textstyle\sum_{i=1}^{2} s_i^p)^{1/p} = (s_1^p + s_2^p)^{1/p}$$
$$\Rightarrow \quad d(u, v) = (u^p + v^p)^{1/p} = (v^p + u^p)^{1/p} = d(v, u). \quad \square \tag{27}$$

Theorem III.8. *Non-negativity:*

$$d(s_1, s_2) = (\textstyle\sum_{i=1}^{2} s_i^p)^{1/p}, \quad p \in N \quad \wedge \quad s_1, s_2 \geq 0 \quad \Rightarrow \quad d(u, v) \geq 0.$$

Proof.

$$\forall\, n \geq 1 \quad \wedge \quad s_1, s_2 \geq 0 \quad \Rightarrow \quad d(s_1, s_2) = (s_1^p + s_1^p)^{1/p} \geq 0. \tag{28}$$

$$\forall\, u = s_1,\ v = s_2 \ \wedge\ d(s_1, s_2) = (s_1^p + s_1^p)^{1/p} \geq 0 \ \Rightarrow\ d(u, v) = (u^p + v^p)^{1/p} \geq 0. \quad \square \tag{29}$$

Theorem III.9. *Identity of Indiscernibles:* $d(u, u) = 0$.

Proof. From the non-negativity property (III.8):

$$d(u, w) \geq 0 \quad \wedge \quad d(u, v) \geq 0 \quad \wedge \quad d(v, w) \geq 0 \tag{30}$$

$$\Rightarrow \quad \exists\, d(u, w) = d(u, v) = d(v, w) = 0. \tag{31}$$

$$d(u, w) = d(v, w) = 0 \quad \Rightarrow \quad u = v. \tag{32}$$

$$d(u, v) = 0 \quad \wedge \quad u = v \quad \Rightarrow \quad d(u, u) = 0. \quad \square \tag{33}$$

IV. PROPERTIES LIMITING A SET TO AT MOST 3 ELEMENTS

The following definitions and proof use first order logic. A Horn clause-like expression is used, here, to make the proof easier to read. By convention, the proof goal, in a Horn clause, is on the left side of the implication sign ($\leftarrow$) and supporting facts are on the right side. The formal proofs in the Rocq file threed.v are: adj111, adj122, adj212, adj123, adj133, adj233, adj213, adj313, adj323, and not_all_mutually_adjacent_gt_3.

Definition IV.1. Immediate Cyclic Successor of m is n:

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} :$$
$$Successor(m, n, setsize) \quad \leftarrow \quad (m = setsize \wedge n = 1) \quad \vee \quad (n = m + 1 \leq setsize). \tag{34}$$

Definition IV.2. Immediate Cyclic Predecessor of m is n:

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} :$$
$$Predecessor(m, n, setsize) \quad \leftarrow \quad (m = 1 \wedge n = setsize) \quad \vee \quad (n = m - 1 \geq 1). \tag{35}$$

Definition IV.3. Adjacent: Element m is sequentially adjacent to element n if the immediate cyclic successor of m is n or the immediate cyclic predecessor of m is n. Notionally:

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} :$$
$$Adjacent(m, n, setsize) \leftarrow Successor(m, n, setsize) \vee Predecessor(m, n, setsize). \tag{36}$$

Definition IV.4. Immediate Symmetric Cylic List (ISCL), where every set element is sequentially adjacent to every other element):

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} : \quad Adjacent(m, n, setsize). \tag{37}$$

Theorem IV.5. *An ISCL is limited to at most 3 elements.*

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} : \quad Adjacent(m, n, setsize) \quad \Rightarrow \quad setsize \leq 3. \tag{38}$$

Proof.

Every element is adjacent to every other element, where $setsize \in \{1,\ 2,\ 3\}$:

$$Adjacent(1,1,1) \leftarrow Successor(1,1,1) \leftarrow (m = setsize \wedge n = 1). \tag{39}$$

$$Adjacent(1,2,2) \leftarrow Successor(1,2,2) \leftarrow (n = m + 1 \leq setsize). \tag{40}$$

$$Adjacent(1,2,3) \leftarrow Successor(1,2,3) \leftarrow (n = m + 1 \leq setsize). \tag{41}$$

$$Adjacent(2,1,3) \leftarrow Predecessor(2,1,3) \leftarrow (n = m - 1 \geq 1). \tag{42}$$

$$Adjacent(3,1,3) \leftarrow Successor(3,1,3) \leftarrow (n = setsize \wedge m = 1). \tag{43}$$

$$Adjacent(1,3,3) \leftarrow Predecessor(1,3,3) \leftarrow (m = 1 \wedge n = setsize). \tag{44}$$

$$Adjacent(2,3,3) \leftarrow Successor(2,3,3) \leftarrow (n = m + 1 \leq setsize). \tag{45}$$

$$Adjacent(3,2,3) \leftarrow Predecessor(3,2,3) \leftarrow (n = m - 1 \geq 1). \tag{46}$$

Element 2 is the only immediate successor of element 1 for all $setsize \geq 3$, which implies element 3 is not ($\neg$) an immediate successor of element 1 for all $setsize \geq 3$:

$$\neg Successor(1,3, setsize \geq 3) \leftarrow \\ Successor(1,2, setsize \geq 3) \leftarrow (n = m + 1 \leq setsize). \tag{47}$$

Element $n = setsize > 3$ is the only immediate predecessor of element 1, which implies element 3 is not ($\neg$) an immediate predecessor of element 1 for all $setsize > 3$:

$$\neg Predecessor(1,3, setsize \geq 3) \leftarrow Predecessor(1, setsize, setsize > 3) \leftarrow \\ (m = 1 \wedge n = setsize > 3). \tag{48}$$

For all $setsize > 3$, some elements are not ($\neg$) sequentially adjacent to every other element (not immediate symmetric):

$$\neg Adjacent(1,3, setsize > 3) \leftarrow \\ \neg Successor(1,3, setsize > 3) \quad \wedge \quad \neg Predecessor(1,3, setsize > 3). \quad \square \tag{49}$$

The Symmetric goal matches the Adjacent goals 39 and fails for all "setsize" greater than three.

V. DERIVATION OF PHYSICS EQUATIONS AND CONSTANTS

A. The direct and inverse unit ratios

Application of the ISCL property of volume (IV.5) and the Euclidean distance dimension $\subseteq \mathbb{R}$ (III.3):

1. **ISCL:** $\{r_1, r_2, r_3\}$ is an ISCL of 3 unordered "distance" dimensions, each dimension having the $\subseteq \mathbb{R}$ set membership property and $\{t\ (time), m\ (mass),\ q\ (charge)\}$ is the ISCL of 3 unordered "non-distance" dimensions, each dimension $\subseteq \mathbb{R}$. There is a total of 6 unordered dimensions with the $\subseteq \mathbb{R}$ property: $r_1,\ r_2,\ r_3,\ t,\ m,\ q$.

2. **Corresponding lengths:** Where the dimensions of Euclidean distance, time, mass, and charge are each divided into the same number of intervals, the i^{th} unit interval length, r_p of Euclidean distance, corresponds to the i^{th} unit interval lengths: t_p of time, m_p of mass, and q_p of charge, where "unit" means a measurement type, for example, meters, seconds, kilograms, and Coulombs.

There are 3 direct proportion ratios from the properties of $\mathbb{R}$:

$$\forall\ r_p, t_p, m_p, q_p, k \in \mathbb{R},\ \exists\ r, t, m, q \in \mathbb{R}:$$
$$\frac{r_p}{t_p} = \frac{r_p k}{t_p k} = \frac{r}{t} \quad \wedge \quad \frac{r_p}{m_p} = \frac{r_p k}{m_p k} = \frac{r}{m} \quad \wedge \quad \frac{r_p}{q_p} = \frac{r_p k}{q_p k} = \frac{r}{q} \tag{50}$$

$$\exists\ c, c_t, c_m, c_q \in \mathbb{R}: \quad r_p/t_p = c_t = c \quad \wedge \quad r_p/m_p = c_m \quad \wedge \quad r_p/q_p = c_q$$
$$\Rightarrow \quad r/t = r_p/t_p = c_t = c \quad \wedge \quad r/m = r_p/m_p = c_m \quad \wedge \quad r/q = r_p/q_p. = c_q. \tag{51}$$

And there are 3 inverse proportion ratios from the properties of $\mathbb{R}$:

$$\forall\ r_p, t_p, m_p, q_p, k \in \mathbb{R},\ \exists\ r, t, m, q \in \mathbb{R}:$$
$$r_p t_p = r_p t_p \left(\frac{k}{k}\right) = \left(\frac{r_p}{k}\right)(t_p k) = rt \quad \wedge \quad r_p m_p = r_p m_p \left(\frac{k}{k}\right) = \left(\frac{r_p}{k}\right)(m_p k) = rm$$
$$\wedge \quad r_p q_p = r_p q_p \left(\frac{k}{k}\right) = \left(\frac{r_p}{k}\right)(q_p k) = rq \tag{52}$$

$$\exists\ k_t, k_m, k_q \in \mathbb{R}: \quad r_p t_p = k_t \quad \wedge \quad r_p m_p = k_m \quad \wedge \quad r_p q_p = k_q$$
$$\Rightarrow \quad rt = t_p r_p = k_t \quad \wedge \quad rm = m_p r_p = k_m \quad \wedge \quad rq = q_p r_p = k_q. \tag{53}$$

B. G, Newton, Gauss, Poisson gravity laws

From equations 51, linear acceleration, r/t^2, can be related to mass, m, and distance, r:

$$r = c_m m \ \wedge \ r = c_t t \ \Rightarrow \ \frac{r}{(c_t t)^2} = \frac{c_m m}{r^2} \ \Rightarrow \ \frac{r}{t^2} = \frac{(c_m c_t^2) m}{r^2} = \frac{Gm}{r^2}, \tag{54}$$

where $G = c_m c_t^2 = r_p^3 / m_p t_p^2$ conforms to the SI units: $m^3 \cdot kg^{-1} \cdot s^{-2}$.[12]

Newton's law follows from multiplying both sides of equation 54 by m:

$$\frac{r}{t^2} = \frac{Gm}{r^2} \ \Leftrightarrow \ F \overset{\text{def}}{=} \frac{mr}{t^2} = \frac{Gm^2}{r^2}. \tag{55}$$

$$F = \frac{Gm^2}{r^2} \ \wedge \ \forall\, m \in \mathbb{R} : \ \exists\, m_1, m_2 \in \mathbb{R} : \ m_1 m_2 = m^2 \quad \Rightarrow \quad F = \frac{G m_1 m_2}{r^2}. \tag{56}$$

A new interpretation of Gauss's and Poisson's gravity laws is presented here. Equation 54 relates linear acceleration, r/t^2, to mass and distance. Gauss's gravity field, $\mathbf{g}$, and Poisson's gravity field, $-\nabla\Phi(\tilde{\mathbf{r}}, t)$, is the angular acceleration, $2\pi r/t^2$. Applying equation 54:

$$\frac{r}{t^2} = -\frac{Gm}{|\tilde{\mathbf{r}}|^2} \ \wedge \ \mathbf{g} = -\nabla\Phi(\tilde{\mathbf{r}}, t) = \frac{2\pi r}{t^2} \ \Rightarrow \ \mathbf{g} = -\nabla\Phi(\tilde{\mathbf{r}}, t) = -\frac{2\pi G m}{|\tilde{\mathbf{r}}|^2}. \tag{57}$$

$$\mathbf{g} = -\nabla\Phi(\tilde{\mathbf{r}}, t) = -\frac{2\pi G m}{|\tilde{\mathbf{r}}|^2} \quad \Rightarrow \quad \nabla \cdot \mathbf{g} = \nabla^2\Phi(\tilde{\mathbf{r}}, t) = \frac{4\pi G m}{|\tilde{\mathbf{r}}|^3} = 4\pi G \rho, \tag{58}$$

where $\rho = m/|\tilde{\mathbf{r}}|^3$ is the mass density:

C. Special relativity

$$\forall\, \tau \in \{t, m, q\},\ r^2 = r'^2 + r_v^2,\ \exists\, \mu, \nu : \ r = \mu\tau \quad \wedge \quad r_v = \nu\tau \quad \Rightarrow \quad (\mu\tau)^2 = r'^2 + (\nu\tau)^2$$
$$\Rightarrow \quad r' = \sqrt{(\mu\tau)^2 - (\nu\tau)^2} = \mu\tau\sqrt{1 - (\nu/\mu)^2}. \tag{59}$$

Local frame distance, r', contracts relative to a distant observer frame distance, r, as $\nu \to \mu$:

$$r' = \mu\tau\sqrt{1 - (\nu/\mu)^2} \quad \wedge \quad \mu\tau = r \quad \Rightarrow \quad r' = r\sqrt{1 - (\nu/\mu)^2}. \tag{60}$$

A distant observer frame type, τ, dilates relative to the local observer frame type, τ', as $\nu \to \mu$:

$$\mu\tau = r'/\sqrt{1 - (\nu/\mu)^2} \quad \wedge \quad r' = \mu\tau \quad \Rightarrow \quad \tau = \tau'/\sqrt{1 - (\nu/\mu)^2}. \tag{61}$$

Where τ is type, time, the space-like flat Minkowski spacetime event interval is:

$$\mathrm{d}r^2 = \mathrm{d}r'^2 + \mathrm{d}r_v^2 \quad \wedge \quad \mathrm{d}r_v^2 = \mathrm{d}r_1^2 + \mathrm{d}r_2^2 + \mathrm{d}r_3^2 \quad \wedge \quad \mathrm{d}(\mu\tau) = \mathrm{d}r$$
$$\Rightarrow \ \mathrm{d}r'^2 = \mathrm{d}(\mu\tau)^2 - \mathrm{d}r_1^2 - \mathrm{d}r_2^2 - \mathrm{d}r_3^2. \tag{62}$$

D. Exact general relativity solutions

The Einstein field equation (EFE) is:

$$\mathbf{G}_{\mu,\nu} + \Lambda g_{\mu,\nu} = \kappa T_{\mu,\nu}, \quad \mathbf{G}_{\mu,\nu} = \mathbf{R} + \frac{1}{2} R g_{\mu,\nu}. \tag{63}$$

where: $\mathbf{G}_{\mu,\nu}$ is Einstein's tensor, $\mathbf{R}$ is the Ricci curvature, R is the scalar curvature, $g_{\mu,\nu}$ is the metric tensor, Λ is a cosmological constant, $\kappa = 8\pi G/c^4$, is Einstein's constant, and $T_{\mu,\nu}$ is the stress-energy-momentum tensor.[4,5]

The EFE extends the previously derived Poisson's law of gravity (58) to include a time component. The Minkowski spacetime component for time is $c^2 g_{0,0} \mathrm{d}t^2$.

$$\frac{c^2}{2}(\mathbf{G}_{0,0} + \Lambda g_{0,0}) = \nabla^2 \Phi(\tilde{\mathbf{r}}, t) = -4\pi G \rho = -4\pi G \frac{m}{|\mathbf{r}|^3} = -\frac{4\pi G}{c^2} \cdot \frac{mc^2}{|\mathbf{r}|^3} = \frac{4\pi G}{c^2} T_{0,0}. \tag{64}$$

$$\frac{c^2}{2}(\mathbf{G}_{0,0} + \Lambda g_{0,0}) = \frac{4\pi G}{c^2} T_{0,0} \quad \Rightarrow \quad \mathbf{G}_{0,0} + \Lambda g_{0,0} = \frac{8\pi G}{c^4} T_{0,0} = \kappa T_{0,0}. \tag{65}$$

In Einstein's general theory of relativity, the stress-energy tensor, $T_{\mu,\nu}$, is considered the independent variable that "causes" the components of $\mathbf{G}_{\mu,\nu}$ and $g_{\mu,\nu}$. But: 1) The laws of physics are independent of the Ricci curvature and independent of the distribution of stress, energy, and momentum. 2) The laws of physics are specified as the components of the metric, $g_{\mu,\nu}$. Derivation of the Schwarzchild metric, in next subsection, is an example of the following simple steps to derive exact metrics independent of the EFE:

Step 1) Use the unit ratios and equations (laws) derived from the ratios, for example, the ratio-derived special relativity equations (60) and the ratio-derived constant, G (54), to derive the scalar functions (laws) assigned to the components of $g_{\nu,\mu}$.

Step 2) The 4D flat spacetime interval equation is an instance of the 2D equation (62). Therefore, derive the metric first as a 2D metric and expand later to a 4D metric.

(Optional step 2d) The 2D metric tensor allows using the much simpler 2D Ricci curvature (which, in 2D, is the Gaussian curvature). And the 2D tensors reduce the number of independent equations to solve. The 2D solutions can next be used to set constraints on the solutions in the 4D tensor, much like an Euclidean distance value places constraints on the coordinate distance values.

Step 3) Translate from 2D to 4D. For spherically symmetric cases, the simplest method to translate from 2D to 4D is to use spherical coordinates, where the first 2 diagonal components of $g_{\nu,\mu}$ remain unchanged and the other 2 diagonal components are functions of the longitude angle, θ, and latitude angle, ϕ.

E. Schwarzschild's time dilation and metric

From equations 60 and 51:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-(1)(v^2/c^2)} \quad \wedge \quad c_m m/r = 1$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-c_m m v^2/rc^2}. \quad (66)$$

Where v_{escape} is the escape velocity and KE is the kinetic energy:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-c_m m v^2/rc^2} \quad \wedge \quad KE = mv^2/2 = mv^2_{escape}$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m m v^2_{escape}/rc^2}. \quad (67)$$

For a photon, the escape velocity, $v_{escape} = c$.

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m m v^2_{escape}/rc^2}. \quad \wedge \quad v_{escape} = c$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m mc^2/rc^2}. \quad (68)$$

Combining equation 68 with the derivation of G (54):

$$c_m c^2 = G \wedge \sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m mc^2/rc^2} \Rightarrow \sqrt{1-(v^2/c^2)} = \sqrt{1-2Gm/rc^2}. \quad (69)$$

Combining equation 69 with equation 61 yields Schwarzschild's gravitational time dilation:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2Gm/rc^2} \wedge t' = t\sqrt{1-(v^2/c^2)} \Rightarrow t' = t\sqrt{1-2Gm/rc^2}. \quad (70)$$

From equations 60 and 69, where Schwarzschild defined the black hole event horizon radius as $\alpha \overset{\text{def}}{=} 2Gm/c^2$.[14,15]

$$r' = r\sqrt{1-(v/c)^2} = r\sqrt{1-2Gm/rc^2} \quad \wedge \quad \alpha \overset{\text{def}}{=} 2Gm/c^2 \quad \Rightarrow \quad r' = r\sqrt{1-\alpha/r}. \quad (71)$$

Applying equation 71 to the time-like spacetime interval equation 62, where $\mathrm{d}s^2 < 0$:

$$r' = r\sqrt{1-\alpha/r} \quad \wedge \quad \mathrm{d}s^2 = \mathrm{d}r'^2 - \mathrm{d}r^2$$
$$\Rightarrow \quad \mathrm{d}s^2 = (\sqrt{1-\alpha/r}\mathrm{d}r)^2 - (\mathrm{d}r'/\sqrt{1-\alpha/r})^2 = (1-\alpha/r)\mathrm{d}r^2 - (1-\alpha/r)^{-1}\mathrm{d}r'^2. \quad (72)$$
$$\mathrm{d}s^2 = (1-\alpha/r)\mathrm{d}r^2 - (1-\alpha/r)^{-1}\mathrm{d}r'^2 \quad \wedge \quad \mathrm{d}r = \mathrm{d}(ct) \quad \wedge \quad c = 1 \quad \wedge$$
$$\lim_{\mathrm{d}s\to 0} \mathrm{d}r' = \lim_{\mathrm{d}s\to 0} \mathrm{d}r \quad \Rightarrow \quad \lim_{\mathrm{d}s\to 0} \mathrm{d}s^2 = (1-\alpha/r)\mathrm{d}t^2 - (1-\alpha/r)^{-1}\mathrm{d}r^2. \quad (73)$$

Using spherical coordinates to translate from 2D to 4D yields the $+ - --$ form of Schwarzschild's black hole metric:[14,15]

$$\lim_{\mathrm{d}s\to 0} \mathrm{d}s^2 = (1-\alpha/r)\mathrm{d}t^2 - (1-\alpha/r)^{-1}\mathrm{d}r^2$$
$$\Rightarrow \quad \lim_{\mathrm{d}s\to 0} \mathrm{d}s^2 = (1-\alpha/r)\mathrm{d}t^2 - (1-\alpha/r)^{-1}\mathrm{d}r^2 - r^2(\mathrm{d}\theta^2 + sin^2\theta\mathrm{d}\phi^2)$$
$$\Rightarrow \quad g_{\mu,\nu} = diag[(1-\alpha/r),\ -(1-\alpha/r)^{-1},\ -r^2,\ -r^2 sin^2\theta]. \tag{74}$$

F. k_e, ε_0, E, Coulomb and Gauss laws

From equations 51, linear acceleration, r/t^2, can be related to charge, q, and distance, r:

$$r = c_q q \ \wedge \ r = c_t t \ \Rightarrow \ \frac{r}{(c_t t)^2} = \frac{c_q q}{r^2} \ \Rightarrow \ \frac{r}{t^2} = \frac{(c_q c_t^2)q}{r^2}. \tag{75}$$

$$r = c_m m \quad \wedge \quad r = c_q q \quad \Rightarrow \quad m = (c_q/c_m)q. \tag{76}$$

Combining equations 75 and 76 yields Coulombs's law:

$$\frac{r}{t^2} = \frac{(c_q c_t^2)q}{r^2} \ \wedge \ m = (c_q/c_m)q \ \Rightarrow \ F \overset{\text{def}}{=} \frac{mr}{t^2} = \frac{(c_{q^2} c_t^2/c_m)q}{r^2} = \frac{k_e q^2}{r^2}. \tag{77}$$

$k_e = c_q^2 c_t^2/c_m = m_p r_p^3/t_p^2 q_p^2$, conforms to the SI units: $kg \cdot m^3 \cdot s^{-2} \cdot C^{-2} \overset{\text{def}}{=} N \cdot m^2 \cdot C^{-2}$.[13]

$$\forall\, q \in \mathbb{R}\ \exists\, q_1, q_2 \in \mathbb{R}:\ q_1 q_2 = q^2 \quad \wedge \quad F = \frac{k_e q^2}{r^2}. \quad \Rightarrow \quad F = \frac{k_e q_1 q_2}{r^2}.. \tag{78}$$

A new interpretation of Gauss's electric field law is presented here. Equation 75 relates the linear acceleration, r/t^2, to charge and distance. Gauss's electric field, **E**, relates angular acceleration, $2\pi r/t^2$ to charge and distance. Applying equation 77:

$$F_C = \frac{mr}{t^2} = \frac{k_e q^2}{r^2} \quad \Rightarrow \quad \exists\, F_E \in \mathbb{R}:\ F_E = \frac{2\pi mr}{t^2} = \frac{2\pi k_e q^2}{r^2}. \tag{79}$$

$$F_E = \frac{2\pi k_e q^2}{r^2} = q(\frac{2\pi k_e q}{r^2}) \quad \wedge \quad \exists\, E \in R:\ E = \frac{2\pi k_e q}{r^2} \quad \Rightarrow \quad F_E = qE. \tag{80}$$

The electric field, $E = 2\pi k_e q/r^2$, conforms to the SI units $kg \cdot m \cdot s^{-2} \cdot C^{-1} \overset{\text{def}}{=} N \cdot C^{-1}$.

$$\mathbf{E} = \frac{2\pi k_e q}{|\vec{\mathbf{r}}|^2} \quad \Rightarrow \quad \nabla \cdot \mathbf{E} = -\frac{4\pi k_e q}{|\vec{\mathbf{r}}|^3}. \tag{81}$$

$$\nabla \cdot \mathbf{E} = -\frac{4\pi k_e q}{|\vec{\mathbf{r}}|^3} \quad \wedge \quad \varepsilon_0 \overset{\text{def}}{=} \frac{1}{4\pi k_e} \quad \wedge \quad \rho \overset{\text{def}}{=} \frac{q}{|\vec{\mathbf{r}}|^3} \quad \Rightarrow \quad \nabla \cdot \mathbf{E} = -\rho/\varepsilon_0, \tag{82}$$

which is Gauss's electric field law in differential form.[13]

G. B, μ_0, and Lorentz's law

Applying the distance contraction equation, 60, to equation 80, where r' is the distant observer frame of reference and r is local (rest) frame of reference:

$$r' = r/\sqrt{1 - v^2/c^2} \quad \wedge \quad F' = \frac{2\pi k_e q^2}{r'^2} \quad \Rightarrow \quad F = \frac{2\pi k_e q^2(1 - v^2/c^2)}{r^2}. \tag{83}$$

From equation 80:

$$E = \frac{2\pi k_e q}{r^2} \;\wedge\; F = \frac{2\pi k_e q^2(1 - v^2/c^2)}{r^2} \;\Rightarrow\; F = q(E - \frac{(2\pi k_e/c^2)q}{r^2}v^2). \tag{84}$$

$$F = q(E - \frac{(2\pi k_e/c^2)q}{r^2}v^2) \;\wedge\; \exists\, B : \; B = \frac{(2\pi k_e/c^2)vq}{r^2} \;\Rightarrow\; F = q(E - Bv), \tag{85}$$

where the magnetic field, $B = (2\pi k_e/c^2)vq/r^2$, conforms to the base SI units: $kg \cdot s^{-1} \cdot C^{-1}$. $\stackrel{\text{def}}{=}$ $kg \cdot s^{-2} \cdot A^{-1}$ $\stackrel{\text{def}}{=}$ T. And assigning the constant portion to μ_0:

$$B = \frac{(2\pi k_e/c^2)vq}{r^2} \quad \Rightarrow \quad \frac{\partial B}{\partial r} = \frac{(4\pi k_e/c^2)vq}{r^3} = \frac{\mu_0 vq}{r^3}. \tag{86}$$

where $\mu_0 = 4\pi k_e/c^2$ conforms to the SI units $kg \cdot m \cdot C^{-2} \stackrel{\text{def}}{=} kg \cdot m \cdot s^{-2}A^{-2}$.

Translating equation 85 to vector form:

$$F = q(E - Bv) \quad \Rightarrow \quad \mathbf{F} = q(\mathbf{E} - \mathbf{B} \times \vec{\mathbf{v}}). \tag{87}$$

$$\mathbf{B} \times \vec{\mathbf{v}} = -(\vec{\mathbf{v}} \times \mathbf{B})) \quad \wedge \quad \mathbf{F} = q(\mathbf{E} - \mathbf{B} \times \vec{\mathbf{v}}) \quad \Rightarrow \quad \mathbf{F} = q(\mathbf{E} + \vec{\mathbf{v}} \times \mathbf{B}), \tag{88}$$

which is Lorentz law in the local frame of reference.

H. Faraday's law

From the magnetic field equation 88, where the electric and magnetic fields are propagating at the speed, $v = c$:

$$B = \frac{(2\pi k_e/c^2)vq}{r^2} \quad \wedge \quad v = c \quad \wedge \quad r = ct \quad \Rightarrow \quad B = \frac{(2\pi k_e/c^3)q}{t^2}. \tag{89}$$

$$B = \frac{(2\pi k_e/c^3)q}{t^2} \quad \Rightarrow \quad \frac{\partial B}{\partial t} = -\frac{(4\pi k_e/c^3)q}{t^3}. \tag{90}$$

$$\frac{\partial B}{\partial t} = -\frac{(4\pi k_e/c^3)q}{t^3} \quad \wedge \quad r = ct \quad \Rightarrow \quad \frac{\partial B}{\partial t} = -\frac{4\pi k_e q}{r^3}. \tag{91}$$

Faraday's law is the case, where the electric field, $\mathbf{E}$, is a single radial vector, $\mathbf{E}_{\tilde{\mathbf{r}}_1}$ acting on some point, P. Therefore, at point, P, $\mathbf{E}_{\tilde{\mathbf{r}}_2} = 0$ and $\mathbf{E}_{\tilde{\mathbf{r}}_3} = 0$. From the Lorentz equation

88, the magnetic field, $\mathbf{B}$ at point P, is perpendicular to $\mathbf{E}_{\tilde{\mathbf{r}}_1}$ (aligned on $\tilde{\mathbf{r}}_2$). For a constant force, the Lorentz equation makes the electric field, $\mathbf{E}_{\tilde{\mathbf{r}}_1}$, a function of $\mathbf{B}_{\tilde{\mathbf{r}}_2}$. Combining with equation 81:

$$\mathbf{E}_{\tilde{\mathbf{r}}_1} = 2\pi k_e q/|\tilde{\mathbf{r}}_2|^2 = f(\mathbf{B}_{\tilde{\mathbf{r}}_2}) \quad \wedge \quad \mathbf{E}_{\tilde{\mathbf{r}}_2} = 0 \quad \wedge \quad \mathbf{E}_{\tilde{\mathbf{r}}_3} = 0 \quad \Rightarrow \quad \nabla \times \mathbf{E} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial \tilde{\mathbf{r}}_1^2} & \frac{\partial}{\partial \tilde{\mathbf{r}}_2^2} & \frac{\partial}{\partial \tilde{\mathbf{r}}_3^2} \\ \mathbf{E}_{\tilde{\mathbf{r}}_1} & 0 & 0 \end{vmatrix}$$

$$= (\frac{\partial \mathbf{E}_{r_3}}{\partial \tilde{\mathbf{r}}_2^2} - \frac{\partial \mathbf{E}_{r_2}}{\partial \tilde{\mathbf{r}}_3^2})\hat{\mathbf{i}} + (\frac{\partial \mathbf{E}_{r_1}}{\partial \tilde{\mathbf{r}}_3^2} - \frac{\partial \mathbf{E}_{r_3}}{\partial \tilde{\mathbf{r}}_1^2})\hat{\mathbf{j}} + (\frac{\partial \mathbf{E}_{r_2}}{\partial \tilde{\mathbf{r}}_1^2} - \frac{\partial \mathbf{E}_{r_1}}{\partial \tilde{\mathbf{r}}_2^2})\hat{\mathbf{k}}$$

$$= 0\hat{\mathbf{i}} + 0\hat{\mathbf{j}} + (0 - \frac{\partial 2\pi k_e q/|\tilde{\mathbf{r}}_2|^2}{\partial |\tilde{\mathbf{r}}_2|^2})\hat{\mathbf{k}} \quad = \quad \frac{4\pi k_e q}{|\tilde{\mathbf{r}}_2|^3}\hat{\mathbf{k}} \quad \equiv \quad \nabla \times \mathbf{E} = \frac{4\pi k_e q}{|\vec{\mathbf{r}}|^3}. \tag{92}$$

Combining equations 92 and 91 yields Maxwell-Faraday's law:[13]

$$\nabla \times \mathbf{E} = \frac{4\pi k_e q}{|\vec{\mathbf{r}}|^3} \quad \wedge \quad \frac{\partial \mathbf{B}}{\partial t} = -\frac{4\pi k_e q}{|\vec{\mathbf{r}}|^3} \quad \Rightarrow \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \tag{93}$$

I. Maxwell's wave equations

From the Gauss's electric field equations 82 and 86:

$$\mathbf{E} = \frac{2\pi k_e q}{|\vec{\mathbf{r}}|^2} \quad \Rightarrow \quad \nabla^2 \mathbf{E} = \frac{12\pi k_e q}{|\vec{\mathbf{r}}|^4}. \tag{94}$$

$$\mathbf{E} = \frac{2\pi k_e q}{|\vec{\mathbf{r}}|^2} \quad \wedge \quad r = ct \quad \Rightarrow \quad \mathbf{E} = \frac{2\pi k_e q}{(ct)^2} \quad \Rightarrow \quad \frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{(12\pi k_e/c^2)q}{t^4}. \tag{95}$$

$$\frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{(12\pi k_e/c^2)q}{t^4} \quad \wedge \quad r = ct \quad \Rightarrow \quad \frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{12\pi k_e c^2 q}{|\vec{\mathbf{r}}|^4}. \tag{96}$$

$$\nabla^2 \mathbf{E} = \frac{12\pi k_e q}{|\vec{\mathbf{r}}|^4} \quad \wedge \quad \frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{12\pi k_e c^2 q}{|\vec{\mathbf{r}}|^4} \quad \Rightarrow \quad \nabla^2 \mathbf{E} = \frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2}. \tag{97}$$

$$\nabla^2 \mathbf{E} = \frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2} \quad \wedge \quad \varepsilon_0\mu_0 = \frac{1}{4\pi k_e}\cdot\frac{4\pi k_e}{c^2} = \frac{1}{c^2} \quad \Rightarrow \quad \nabla^2 \mathbf{E} = \varepsilon_0\mu_0\frac{\partial^2 \mathbf{E}}{\partial t^2}, \tag{98}$$

which is Maxwell's electric field wave equation. And from equation (91):

$$\frac{\partial B}{\partial t} = -\frac{4\pi k_e q}{r^3} \quad \wedge \quad r = |\vec{\mathbf{r}}| \quad \Rightarrow \quad \nabla^2 \mathbf{B} = \frac{12\pi k_e q}{|\vec{\mathbf{r}}|^4}. \tag{99}$$

$$\mathbf{B} = \frac{2\pi k_e q}{|\vec{\mathbf{r}}|^2} \quad \wedge \quad r = ct \quad \Rightarrow \quad \mathbf{B} = \frac{2\pi k_e q}{(ct)^2} \quad \Rightarrow \quad \frac{\partial^2 \mathbf{B}}{\partial t^2} = \frac{(12\pi k_e/c^2)q}{t^4}. \tag{100}$$

$$\frac{\partial^2 \mathbf{B}}{\partial t^2} = \frac{(12\pi k_e/c^2)q}{t^4} \quad \wedge \quad r = ct \quad \Rightarrow \quad \frac{\partial^2 \mathbf{B}}{\partial t^2} = \frac{12\pi k_e c^2 q}{|\vec{\mathbf{r}}|^4}. \tag{101}$$

$$\nabla^2 \mathbf{B} = -12\pi k_e q/|\vec{\mathbf{r}}|^4 \quad \wedge \quad \frac{\partial^2 \mathbf{B}}{\partial t^2} = \frac{12\pi k_e c^2 q}{|\vec{\mathbf{r}}|^4} \quad \Rightarrow \quad \nabla^2 \mathbf{B} = \frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2}. \tag{102}$$

$$\nabla^2 \mathbf{B} = \frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2} \quad \wedge \quad \varepsilon_0\mu_0 = \frac{1}{4\pi k_e}\cdot\frac{4\pi k_e}{c^2} = \frac{1}{c^2} \quad \Rightarrow \quad \nabla^2 \mathbf{B} = \varepsilon_0\mu_0\frac{\partial^2 \mathbf{B}}{\partial t^2}, \tag{103}$$

which is Maxwell's magnetic field wave equation.

J. $\hbar$, h, and Planck-Einstein relation

Applying both the direct proportion ratio (51), and inverse proportion ratio (53):

$$m = k_m/r \quad \wedge \quad r = ct \quad \Rightarrow \quad m(ct)^2 = (k_m/r)r^2 = k_m r. \tag{104}$$

$$m(ct)^2 = k_m r \quad \Rightarrow \quad E \stackrel{\text{def}}{=} mc^2 = k_m r/t^2. \tag{105}$$

$$E = mc^2 = k_m r/t^2 \;\wedge\; r/t = c \;\Rightarrow\; E = mc^2 = k_m c/t. \tag{106}$$

$$E = k_m c/t\ kg\ m^2\ s^{-2} \;\wedge\; k_m c = \hbar\ kg\ m^2\ s^{-1} \;\Rightarrow\; E = \hbar/t\ kg\ m^2\ s^{-2} = \hbar/t\ J. \tag{107}$$

$$E = \hbar/t\ kg\ m^2\ s^{-2} \quad \wedge \quad \omega = (2\pi/t)\ radian\ s^{-1} \quad \Rightarrow$$

$$E_\omega = \hbar(2\pi/t) = \hbar\omega\ kg\ m^2\ radian\ s^{-2}. \tag{108}$$

$$E_\omega = \hbar/t\ kg\ m^2\ radian\ s^{-2} \quad \wedge$$

$$f = (1/t)\ cycle\ s^{-1} = ((2\pi/t)\ radian\ s^{-1})/(2\pi\ radian\ cycle^{-1})$$

$$\Rightarrow \quad E_f = \hbar\omega(2\pi/2\pi) = 2\pi\hbar f = hf\ kg\ m^2\ cycle\ s^{-2} \stackrel{\text{def}}{=} hf\ kg\ m^2\ Hz\ s^{-1}. \tag{109}$$

K. Values of the unit ratios and Planck units

Values of the direct proportion ratios, c_t, c_m, and c_q:

$$c_t = c \approx 2.99792458 \cdot 10^8\ m\ s^{-1}. \tag{110}$$

$$G = c_m c_t^2 \quad \wedge \quad G \approx 6.67430(15) \cdot 10^{-11}\ m^3\ kg^{-1}\ s^{-2} \quad \Rightarrow$$

$$c_m = G/c_t^2 \approx 7.426162 \cdot 10^{-28}\ m\ kg^{-1}. \tag{111}$$

$$k_e = c_q^2 c_t^2/c_m \quad \wedge \quad k_e \approx 8.9875517923(13) \cdot 10^9 Nm^2\ C^{-2} \quad \Rightarrow$$

$$c_q = \sqrt{k_e c_m/c_t^2} \approx 8.6175182 \cdot 10^{-18} m\ C^{-1}. \tag{112}$$

From equation 107, the values of the inverse proportion ratios, k_t, k_m, and k_q:

$$\hbar \approx .054571783 \cdot 10^{-34}\ kg\ m^2\ s^{-1} \;\wedge\; k_m c = \hbar \;\wedge\; c = 299792458\ m\ s^{-1} \;\Rightarrow$$

$$k_m = \hbar/c \approx 3.51767294 \cdot 10^{-43}\ kg\ m. \tag{113}$$

$$r_p = \sqrt{r_p^2} = \sqrt{k_m c_m} \approx 1.616255 \cdot 10^{-35}\ m. \quad \Rightarrow \tag{114}$$

$$k_t = r_p^2/c_t \approx 8.71363 \cdot 10^{-79}\ s\ m. \tag{115}$$

$$k_q = r_p^2/c_q \approx 3.031361 \cdot 10^{-53}\ C\ m. \tag{116}$$

Values of the Planck units, r_p, t_p, m_p, and q_p::

$$r_p = \sqrt{c_t k_t} = \sqrt{c_m k_m} = \sqrt{c_q k_q} \approx 1.616255 \cdot 10^{-35}\ m. \tag{117}$$

$$t_p = r_p/c_t \approx 5.391246 \cdot 10^{-44}\ s. \tag{118}$$

$$m_p = r_p/c_m \approx 2.176434 \cdot 10^{-8}\ kg. \tag{119}$$

$$q_p = r_p/c_q \approx 1.875546 \cdot 10^{-18}\ C. \tag{120}$$

L. The fine structure constant, α

New, simpler equations of the fine structure constants are derived here, where the fine structure is the ratio of an electron's particle's static force to the force of an electron field moving at the speed of light, which reduces to the ratio of the square of the stationary and moving field forces:

$$\alpha_\tau = \frac{F_{\tau_e}}{F_{\tau_p}} = \frac{K_\tau \tau_e^2/r^2}{K_\tau \tau_p^2/r^2} = \frac{\tau_e^2}{\tau_p^2}, \quad \tau \in \{m, q\}. \tag{121}$$

For example, the fine structure electron charge coupling constant is the ratio of stationary and moving Coulomb charge (eletromagnetic) wave forces, which simplifies to:

$$\alpha = \frac{k_e q_e^2/r^2}{k_e q_p^2/r^2} = \frac{q_e^2}{q_p^2} \approx 0.0072973526, \tag{122}$$

where q_e is the elementary (electron) charge and q_p is the Planck charge unit.

The fine structure electron gravity coupling constant, α_G, is the ratio of stationary to moving Newton gravity wave forces, whic simplifies to:

$$\alpha_G = \frac{G m_e^2/r^2}{G m_p^2/r^2} = \frac{m_e^2}{m_p^2} \approx 1.751809 \cdot 10^{-45}, \tag{123}$$

where m_e is the electron mass and m_p is the Planck mass unit.

M. Compton wavelength, λ

[13,16] From equations 53 and 107:

$$rm = k_m \quad \wedge \quad \lambda = 2\pi r \quad \Rightarrow \quad \lambda = 2\pi r = \frac{2\pi k_m}{m} = \frac{2\pi k_m c}{mc} = \frac{2\pi \hbar}{mc} = \frac{h}{mc}. \tag{124}$$

N. Schrödenger's wave equation

Start with the previous unit ratio-derived "base" Planck-Einstein relation 107, $V(r,t)$ isthe potential energy, and multiply the kinetic energy component by mc/mc:

$$mc^2 = \hbar/t \quad \Rightarrow \quad \exists\, V(r,t) :\ \hbar/t = \hbar/2t + V(r,t). \tag{125}$$

$$\hbar/t = h/2t + V(r,t) \quad \Rightarrow \quad \hbar/t = \hbar mc/2mct + V(r,t). \tag{126}$$

And from the distance-to-time (speed of light) ratio (51):

$$\hbar/t = \hbar mc/2mct + V(r,t) \quad \wedge \quad r = ct \quad \Rightarrow \quad \hbar/t = \hbar mc^2/2mcr + V(r,t). \tag{127}$$

$$\hbar/t = \hbar mc^2/2mcr + V(r,t) \ \wedge \ \hbar/t = mc^2 \quad \Rightarrow \quad \hbar/t = \hbar^2/2mcrt + V(r,t). \tag{128}$$

$$\hbar/t = \hbar^2/2mcrt + V(r,t) \quad \wedge \quad r = ct \quad \Rightarrow \quad \hbar/t = \hbar^2/2mr^2 + V(r,t). \tag{129}$$

Multiply both sides of equation 129 by a probability density function, $\Psi(r,t)$.

$$\hbar/t = \hbar^2/2mr^2 + V(r,t) \quad \Rightarrow \quad (\hbar/t)\Psi(r,t) = (\hbar^2/2mr^2)\Psi(r,t) + V(r,t)\Psi(r,t). \tag{130}$$

$$\begin{gathered}(\hbar/t)\Psi(r,t) = ((\hbar)^2/2mr^2)\Psi(r,t) + V(r,t)\Psi(r,t) \quad \wedge \\ \forall\ \Psi(r,t) :\ \partial^2\Psi(r,t)/\partial r^2 = (-1/r^2)\Psi(r,t) \quad \wedge \quad \partial\Psi(r,t)/\partial t = (i\ 1/t)\Psi(r,t) \\ \Rightarrow \quad i\hbar\partial\Psi(r,t)/\partial t = -(\hbar^2/2m)\partial^2\Psi(r,t)/\partial r^2 + V(r,t)\Psi(r,t),\end{gathered} \tag{131}$$

which is the one-dimensional position-space Schrödenger's equation.[6,16]

$$\begin{gathered}i\hbar\partial\Psi(r,t)/\partial t = -(\hbar^2/2m)\partial^2\Psi(r,t)/\partial r^2 + V(r,t)\Psi(r,t) \quad \wedge \quad |\vec{\mathbf{r}}| = r \\ \Rightarrow \quad \exists\, \vec{\mathbf{r}} :\ i\hbar\partial\Psi(\vec{\mathbf{r}},t)/\partial t = -(\hbar^2/2m)\partial^2\Psi(\vec{\mathbf{r}},t)/\partial\vec{\mathbf{r}}^2 + V(\vec{\mathbf{r}},t)\Psi(\vec{\mathbf{r}},t),\end{gathered} \tag{132}$$

which is the 3-dimensional position-space Schrödenger's equation.[6,16]

O. Dirac's wave equation

Start with the previous unit ratio-derived "base" Planck-Einstein relation 107:

$$\hbar/t = mc^2 \quad \Leftrightarrow \quad \hbar/2t + \hbar/2t. = mc^2 \quad \Leftrightarrow \quad \hbar/2t = -\hbar/2t. + mc^2. \tag{133}$$

$$\hbar/2t = -\hbar/2t. + mc^2 \ \wedge \ qc_q/r = qc_q/ct = 1 \ \Rightarrow \ \hbar qc_q/2ct^2 = -qc_q/2rt. + mc^2. \tag{134}$$

$$\hbar qc_q/2ct^2 = -qc_q/2rt. + mc^2 \quad \wedge \quad r = ct \quad \Rightarrow \quad \hbar qc_q/2ct^2 = -qcc_q/2r^2. + mc^2. \tag{135}$$

$$\hbar qc_q/2ct^2 = -qcc_q/2r^2. + mc^2 \quad \Rightarrow \quad \hbar(qc_q/2ct^2)\boldsymbol{\psi} = (-qcc_q/2r^2)\boldsymbol{\psi} + mc^2\boldsymbol{\psi}. \tag{136}$$

$$\psi = e^{-iqc_q((1/2ct)-(1/2r))} \Rightarrow \partial\boldsymbol{\psi}/\partial t = (-iqc_q/2ct^2)\boldsymbol{\psi} \ \wedge \ \nabla\boldsymbol{\psi} = (iqc_q/2r^2)\boldsymbol{\psi}. \tag{137}$$

$$\begin{aligned} &\partial\boldsymbol{\psi}/\partial t = (-iqc_q/2ct^2)\boldsymbol{\psi} \quad \wedge \quad \nabla\boldsymbol{\psi} = (iqc_q/2r^2)\boldsymbol{\psi} \quad \wedge \\ &\hbar(qc_q/2ct^2)\boldsymbol{\psi} = (-qcc_q/2r^2)\boldsymbol{\psi} + mc^2\boldsymbol{\psi} \quad \Rightarrow \quad i\hbar\partial\boldsymbol{\psi}/\partial t = -i\hbar c\nabla\boldsymbol{\psi} + mc^2\boldsymbol{\psi}. \end{aligned} \tag{138}$$

$$\begin{aligned} &\forall\, y = ax + b \ \exists\, f, \boldsymbol{\alpha}, \beta : \ yf(\boldsymbol{\psi}) = \boldsymbol{\alpha}\cdot ax\boldsymbol{\psi} + b\beta\boldsymbol{\psi} \quad \wedge \quad i\hbar\partial\boldsymbol{\psi}/\partial t = -i\hbar c\nabla\boldsymbol{\psi} + mc^2\boldsymbol{\psi} \\ &\Rightarrow \quad \exists\, f, \boldsymbol{\alpha}, \beta : \ i\hbar\partial\boldsymbol{\psi}/\partial t = i\hbar c\boldsymbol{\alpha}\cdot -\nabla\boldsymbol{\psi} + mc^2\beta\boldsymbol{\psi}, \end{aligned} \tag{139}$$

which is the relativistic, free particle Dirac equation[9], where $\beta = \gamma^0$, $\boldsymbol{\alpha} = \gamma^0\gamma^k$, $k \in \{1, 2, 3\}$, and each γ a matrix.

P. Total of a type

Applying both the direct (51) and inverse proportion ratios (53) to the Euclidean distance:

$$\begin{aligned} &r = \sqrt{r_1^2 + r_2^2}, \quad r_1 = c_\tau\tau, \quad r_2 = k_\tau/\tau, \quad \tau \in \{t, m, q\} \\ &\Rightarrow \quad r = \sqrt{(c_\tau\tau)^2 + (k_\tau/\tau)^2} \quad \Leftrightarrow \quad \tau = \sqrt{(r/c_\tau)^2 + (k_\tau/r)^2}. \end{aligned} \tag{140}$$

Q. Quantum extension to Schwarzschild's metric

The simplest way to demonstrate how to add quantum physics to general relativity is by extending Schwarzschild's gravitational time dilation equation and black hole metric. Apply the total of a type equation 140 to the derivation of Schwarzschild's time dilation and metric

(66):

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-(v^2/c^2)(r/r)} \quad \wedge \quad r = \sqrt{(c_m m)^2 + (k_m/m)^2} = Q_m$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-Q_m v^2/rc^2}. \tag{141}$$
$$\sqrt{1-(v^2/c^2)} = \sqrt{1-Q_m v^2/rc^2} \quad \wedge \quad KE = mv^2/2 = mv^2_{escape}$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2Q_m v^2_{escape}/rc^2}. \tag{142}$$

For a photon, the escape velocity, $v_{escape} = c$.

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2Q_m v^2_{escape}/rc^2}. \wedge\ v_{escape} = c$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2Q_m c^2/rc^2} = \sqrt{1-2Q_m/r}. \tag{143}$$

Combining equation 143 with equation 61 yields Schwarzschild's gravitational time dilation with a quantum mass effect:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2Q_m/r} \quad \wedge \quad t' = t\sqrt{1-(v^2/c^2)} \quad \Rightarrow \quad t' = t\sqrt{1-2Q_m/r}. \tag{144}$$

Schwarzschild defined the black hole event horizon radius, $\alpha \stackrel{\text{def}}{=} 2Gm/c^2$. The radius with the quantum extension is $\alpha \stackrel{\text{def}}{=} 2Q_m$. At this point the exact same equations 71 through 74 yield what looks like the same Schwarzschild black hole metric.

R. Quantum extension to Newton's gravity force

The quantum mass effect is easier to understand in the context Newton's gravity equation than in general relativity, because the metric equations and solutions in the EFEs are much more complex. From equations 140 and 51:

$$m/\sqrt{(r/c_m)^2 + (k_m/r)^2} = 1 \quad \wedge \quad r^2/(ct)^2 = 1 \quad \Rightarrow \quad r^2/(ct)^2 = m/\sqrt{(r/c_m)^2 + (k_m/r)^2}$$
$$\Rightarrow \quad r^2/t^2 = c^2 m/\sqrt{(r/c_m)^2 + (k_m/r)^2}. \tag{145}$$
$$r^2/t^2 = c^2 m/\sqrt{(r/c_m)^2 + (k_m/r)^2} \quad \Rightarrow \quad (m/r)(r^2/t^2) = (m/r)(c^2 m/\sqrt{(r/c_m)^2 + (k_m/r)^2})$$
$$\Rightarrow \quad F \stackrel{\text{def}}{=} mr/t^2 = c^2 m^2/\sqrt{(r^4/c_m^2) + k_m^2}. \tag{146}$$
$$F = c^2 m^2/\sqrt{(r^4/c_m^2) + k_m^2} \quad \wedge \quad \forall\, m \in \mathbb{R},\ \exists\, m_1, m_2 \in \mathbb{R} :\ m_1 m_2 = m^2$$
$$\Rightarrow \quad F = c^2 m_1 m_2/\sqrt{(r^4/c_m^2) + k_m^2}. \tag{147}$$

S. Quantum extension to Coulomb's charge force

$$q^2/((r/c_q)^2+(k_q/r)^2)=1\ \wedge\ r^2/(ct)^2=1\ \Rightarrow\ r^2/(ct)^2=q^2/((r/c_q)^2+(k_q/r)^2)$$
$$\Rightarrow\quad r^2/t^2=c^2q^2/((r/c_q)^2+(k_q/r)^2). \tag{148}$$
$$r^2/t^2=c^2q^2/((r/c_q)^2+(k_q/r)^2)\quad\wedge\quad r=c_m m$$
$$\Rightarrow\quad F\stackrel{\text{def}}{=}mr/t^2=(c^2/c_m)q^2/((r/c_q)^2+(k_q/r)^2). \tag{149}$$
$$\forall\, q\in\mathbb{R}:\ \exists\, q_1,q_2\in\mathbb{R}:\ q_1q_2=q^2\quad\wedge\quad F=(c^2/c_m)q^2/((r/c_q)^2+(k_q/r)^2)$$
$$\Rightarrow\quad F=(c^2/c_m)q^2/((r/c_q)^2+(k_q/r)^2). \tag{150}$$

VI. INSIGHTS AND IMPLICATIONS

1. A distance measure, d, is an inverse (bijective) function of a volume, v, where v is the sum of subset volumes (Euclidean or non-Euclidean), $g(s_{i,1},\cdots,s_{i,n})$. Notionally:

$$\forall\, n\in\mathbb{N},\ \ f,g:\ d\geq 0,\ \ d=f^{-1}(v)=f^{-1}(\textstyle\sum_{i=1}^m g(s_{i,1},\cdots,s_{i,n})): \tag{151}$$

   (a) shows the intimate relation between distance and volume that definitions, like inner product space and metric space, ignore.[2,3]

   (b) is a more simple and concise definition of a distance measure, having the metric space properties.[2,3]

2. The left side of the distance sum inequality (III.5),

$$(\textstyle\sum_{i=1}^m(a_i^n+b_i^n))^{1/n}\leq(\sum_{i=1}^m a_i^n)^{1/n}+(\sum_{i=1}^m b_i^n)^{1/n}, \tag{152}$$

   differs from the left side of Minkowski's sum inequality.[11]

$$(\textstyle\sum_{i=1}^m(a_i^n+b_i^n)^{\mathbf{n}})^{1/n}\leq(\sum_{i=1}^m a_i^n)^{1/n}+(\sum_{i=1}^m b_i^n)^{1/n}. \tag{153}$$

   (a) The two inequalities are only the same where $n=1$.

   (b) The distance sum inequality (III.5) is a more fundamental inequality because the proof does not require the convexity and Hölder's inequality assumptions required to prove the Minkowski sum inequality.[11]

(c) $\forall\ n > 1,\ v > 0$: The distance sum inequality term, $v_i^n = a_i^n + b_i^n$: $d = v^{1/n} = (\sum_{i=1}^m v_i^n)^{1/n}$, is the Minkowski distance, which makes the distance sum inequality directly related to geometry. But the Minkowski sum inequality term, $d = v^{1/n} = (\sum_{i=1}^m ((v_i^n)^{\mathbf{n}}))^{1/n} = (\sum_{i=1}^m v_i^{\mathbf{n^2}})^{1/n}$, is *not* a Minkowski distance.

(d) The distance sum inequality might be useful for measuring distance between concepts in machine learning.

3. *Combinatorics.* The number of n-tuples, $v_c = \prod_{i=1}^n |x_i|$, was proven to imply the Euclidean volume equation (II.1). And the notion of distance was derived as the inverse function of a volume. The equations were derived without relying on the geometric primitives and relations in Euclidean geometry[17,18], axiomatic geometry[19–23], trigonometry[24,25], calculus[24,26,27], and vector analysis.[5]

4. *Combinatorics.* The commutative property of the operations defining volume and distance equations requires being able to sequence a set of n number of unordered domain elements in $n!$ number of permuted sequences, which limits the set to $n \leq$ elements (IV.5). Other elements must have different types (are elements of different sets).

   (a) If there are more than 3 distance and 3 non-distance dimensions $\subseteq \mathbb{R}$, then the additional dimensions $\subseteq \mathbb{R}$ are ordered, where each dimension has an intrinsic property having a fixed position in strict totally ordered set.

   (b) If a set of quantum states is unordered, then there are at most 3 states of the same type.

5. A discrete (point) valued state has measure 0 (zero-length interval). The ratio of a time, distance, mass, or charge interval length to zero is undefined, which is the reason discrete, state values exist independent of distance (position), time, mass, and charge. The value from a set of unordered, discrete state values is undetermined with respect to space and time until observed.

6. For each unit interval length, r_p, of a 3-dimensional distance, there are unit interval lengths of 3 other types of dimensions, forming 3 direct proportion unit ratios (51): $r = (r_p/t_p)t = (r_p/m_p)m = (r_p/q_p)q$ and 3 inverse proportion ratios (53): $r = (r_pt_p)/t = (r_pm_p)/m = (r_pq_p)/q$, where r_p, t_p, m_p, and q_p are the Planck units (117).

7. In this article, the following constants are derived directly from the ratios and Planck units: gravity, $G = c_m c_t^2 = (r_p/m_p)(r_p/t_p)^2 = r_p^3/m_p t_p^2$ (56), reduced Planck $\hbar = k_m c_t = (r_p m_p)(r_p/t_p) = r_p^2 m_p/t_p$ (107), fine-structure $\alpha = q_e^2/q_p^2$ (123), charge, $k_e = c_q^2 c_t^2/c_m$ (78), vacuum permittivity $\varepsilon_0 = 1/4\pi k_e = 1/4\pi(c_q^2 c_t^2/c_m)$ (82), and vacuum permeability $\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m = 4\pi r_p m_p/q_p^2$ (86).

   And the quantum extensions to: Schwarzschild's gravitational time dilation (143), Newton's gravity force (147), and Coulomb's charge force (150), show that the constants G, $\kappa = 8\pi G/c^4$, k_e, ε_0, and μ_0 do not exist (not valid), where the quantum effects are measurable.

   Therefore, **none** of the following constants: G, κ, $\hbar$, $h = 2\pi\hbar$, α, k_e, ε_0, and μ_0 are fundamental constants because 1) they were all shown to be derived from the ratios and Planck units and 2) G, κ, k_e, ε_0, and μ_0 do not exist where the quantum effects are measurable.

8. An empirical law *describes* the relations between the variables in an equation. Deriving empirical laws from the ratios *explains* the reasons for relations between the variables in an equation. Further, all the derivations of the physics equations from the ratios are much shorter and simpler than other derivations, which shows that the ratios are an important tool for scientists and engineers.

9. c_t, c_m, and c_q are the *maximum* ratios: $r = \sqrt{r_1^2 + r_2^2 + r_3^2} \;\Rightarrow\; r_1,\ r_2,\ r_3 \le r$ $\Rightarrow\ \forall\, \tau \in \{t, m, q\} : r_1/\tau,\ r_2/\tau,\ r_3/\tau \le r/\tau = r_p/\tau_p = c_\tau$. For example, the speeds of gravity and light waves are limited to the speed, c_t.

10. c_t is a component of the constants: $G = c_m c_t^2$, $k_e = c_q^2 c_t^2/c_m$, $\varepsilon_0 = 1/4\pi k_e = 1/4\pi(c_q^2 c_t^2/c_m)$, and $\hbar = k_m c_t$. The only constants, derived in this article, that do not contain c_t are: vacuum permeability constant: $\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m$, and the fine structure constants, $\alpha = q_e^2/q_p^2$ and $\alpha_G = m_e^2/m_p^2$.

11. Simpler equations for the fine structure constants were derived, where the fine structure is the ratio of an electron's particle's static force to the force of an electron's wave field force moving at the speed of light, which reduces to the ratio of the square of the stationary and moving field forces: $\alpha_\tau = \tau_e^2/\tau_p^2$, $\tau \in \{m, q\}$.

The ratios (interactions) of two subtypes of a force, is a parsimonious explanation for the fine structure equations. In contrast, the complicated fine structure equations, currently in use, have no simple explanations.

(a) The following steps show that the unnecessarily complicated CODATA definition, $\alpha = q_e^2/4\pi\varepsilon_0\hbar c$, reduces to the simpler equation-derived in this article:

$$\varepsilon_0 = 1/4\pi k_e = 1/(4\pi(c_q^2 c_t^2/c_m) \quad \wedge \quad \hbar = k_m c_t \quad \wedge \quad h = 2\pi\hbar$$
$$\Rightarrow \quad \varepsilon_0 hc = 2\pi k_m c_t^2/(4\pi(c_q^2/c_m)c_t^2) = k_m/(2(c_q^2/c_m))$$
$$= m_p r_p/(2((r_p/q_p)^2/(r_p/m_p))) = q_p^2/2. \quad (154)$$

$$\alpha = q_e^2/2\varepsilon_0 hc \quad \wedge \quad \varepsilon_0 hc = q_p^2/2 \quad \Rightarrow \quad \alpha = q_e^2/2(q_p^2/2) = q_e^2/q_p^2. \quad (155)$$

(b) The unnecessarily complicated fine structure electron gravity coupling constant reduces to the simpler equation derived in this article:

$$\alpha_G = Gm_e^2/\hbar c = (c_m c_t^2)m_e^2/(k_m c_t)c_t = (r_p/m_p)m_e^2/m_p r_p = m_e^2/m_p^2. \quad (156)$$

12. The assumption that the old definition, $\mu_0 = 4\pi k_e/c_t^2 = 1/\varepsilon_0 c^2$, and the newer NIST/CODATA definition, $\mu_0 = 2h\alpha/cq_e^2$,[28] are different because of slightly different values is *incorrect*! The following steps will show that both equations reduce to the same base equation, $\mu_0 = 4\pi k_m/q_p^2$, which shows that the old and newer NIST/CODATA definitions differ in value because of different accuracies of the empirical values, k_e and ε_0, versus the values, h and α. The values of h and α could be used to derive more accurate values of k_e and ε_0.

Using the ratios to reduce the old definition of μ_0 derived in this article:

$$\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m = 4\pi(r_p/q_p)^2/(r_p/m_p) = 4\pi m_p r_p/q_p^2 = 4\pi k_m/q_p^2. \quad (157)$$

Using the ratios to reduce the CODATA definition, $\mu_0 = 2h\alpha/cq_e^2$:

$$h = 2\pi k_m c_t, \;\; \alpha = q_e^q/q_p^2 \;\; \Rightarrow \;\; \mu_0 = 2h\alpha/cq_e^2 = 4\pi k_m c_t(q_e^2/q_p^2)/c_t q_e^2 = 4\pi k_m/q_p^2. \quad (158)$$

13. The ratio-based derivation of Lorentz's law here (88) differs from other relativistic derivations, in that equations for the magnetic field, $\mathbf{B}$, and magnetic permeability, μ_0, are derived in the process. The equation, $\mathbf{B} = (2\pi k_e/c^2)vq/|\vec{\mathbf{r}}|^2$, allows a simple derivation of the Faraday's law (93) and the magnetic component of Maxwell's electromagnetic equations (103).

14. Maxwell's derivation of the electric and magnetic field wave equations required: Gauss's electric field law in a vacuum: $\nabla \cdot \mathbf{E} = -\rho/\varepsilon_0$, Gauss's magnetic field law in a vacuum: $\nabla \cdot \mathbf{B} = 0$, Faraday's Law: $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$, and Ampere's Law: $\nabla \times \mathbf{B} = \mu_0 \varepsilon_0 \partial \mathbf{E}/\partial t$.[13] In contrast, the derivations (98) (103), in this article, did not require any of those laws. And the derivations, in this article, were shorter and simpler.

15. As shown in the subsection deriving the Schwarzschild's gravitational time dilation and black hole metric (66)[14,15] using ratios illustrates a simple way of finding exact solutions to Einstein's field equations.

16. Einstein's relativity equations: 1) assume the Lorentz transformations, 2) assume the laws of physics are same at each coordinate point, 3) assume the notion of light, and 4) assume that the speed of light is the same at each coordinate point.[4,5]

    The derivations, in this article, were made without those assumptions (does not even require the notion of light). The unit ratios are ratios of Euclidean distance to time, mass, and charge. Therefore, the unit ratios (like the maximum speed ratio, c_t), and all equations (laws) and constants derived from those ratios must be the same at each local coordinate point.

17. There are 3 Planck-Einstein relations: $E_\omega = \hbar\omega \; kg \; m^2 \; radian \; s^{-2}$ (108) and $E_f = hf \; kg \; m^2 \; Hz \; s^{-1}$ (109), which are both instances of a base Planck-Einstein relation, $E = mc^2 = \hbar/t \; kg \; m^2 \; s^{-2} = \hbar/t \; J$ (107). That is, the assumption that the value, 1, in the term, $E = \hbar(1/t)$, has units is incorrect.

18. CODATA defines the Planck constant to be an "atomic" constant with a defined empirical value and defines the reduced Planck constant as $\hbar = h/2\pi$.[28] But, it was shown, in this article, that the reduced Planck constant $\hbar = k_m c_t$, where the full Planck constant, $h = 2\pi\hbar$ Therefore, the full and reduce Planck constants are not atomic constants.

19. Derivations of Schrödenger's wave equation (132)[6] and Dirac's wave equation (139)[9] from the base Planck-Einstein relation, which uses the reduced Planck constant, are shorter, simpler, and require fewer assumptions. And derivations of the Planck units from the reduced Planck constant are simpler and shorter.

Therefore, it would be more logical for CODATA to define the reduced Planck constant as, $\hbar = k_m c_t$, define its value, and define the full Planck constant in terms of the reduced Planck constant.

20. The quantum extensions to: Schwarzschild's black hole metric (74), Newton's gravity force (147), and Coulomb's charge force (150) make quantifiable predictions:

    (a) For gravity, $\lim_{r\to 0} F = c^2 m_1 m_2/k_m$, and for charge, $\lim_{r\to 0} F = 0$. Finite maximum gravity and charge forces: 1) eliminates the problem of forces going to infinity as $r \to 0$, 2) might allow radioactivity without the need for a weak force, and 3) finite sloped energy well walls might facilitate quantum tunneling.

    (b) The quantum-extended relativity and classic equations reduce to the non-extended relativity and classic equations and constants, where the distance between 2 masses and between 2 charges is sufficiently large or the masses and charges sufficiently large that the quantum effects are not measurable. *Note* that G, k_e, ε_0, μ_0, and κ (Einstein's constant, which contains G) do not exist (are not valid), where the quantum effects becomes measurable.

    (c) The covariant tensor components, in Einstein's field equations, that had the units $1/distance^2$, will now have the more complex units, $1/\sqrt{(distance^4/c_\tau^2) + k_\tau^2}$, $\tau \in \{t, m\}$.

    (d) $1/\sqrt{(distance^4/c_\tau^2) + k_\tau^2}$ implies that as distance $\to 0$, spacetime curvature peaks at the Planck unit length (distance), r_p (117). The ratio of quantum units, $m_p/r_p^3 \approx 5.15484\dot{8}10^{96}\ kg\ m^{-3}$, might indicate a maximum mass density. A finite force (spacetime curvature) and finite mass density might imply that black holes have sizes > 0 (are not singularities). It might allow black hole evaporation. If there was a "big bang," then it might not have originated from a singularity.

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